```
Frequency Distribution of Customer Ages: Age

22    3
25    2
28    1
30    2
Name: count, dtype: int64
```

```
import pandas as pd #

Sample data

data = {
    "Post": ["P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"], "Likes": [120,
    150, 120, 200, 150, 180, 120, 200]
}

df = pd.DataFrame(data)

# Frequency distribution of likes
frequency = df["Likes"].value_counts().sort_index()

print("Frequency Distribution of Likes:")
print(frequency)
```

```
Frequency Distribution of Likes: Likes

120    3
150    2
180    1
200    2
Name: count, dtype: int64
```

```
# Import libraries
import pandas as pd
from google.colab import files

# Upload any CSV file
uploaded = files.upload()

# Read uploaded file
filename = list(uploaded.keys())[0] df =
pd.read_csv(filename)

# Display dataset
print("\n========== DATASET ==========\n")
print(df)

# Number of rows and columns
print("\n========== DATASET SHAPE ==========")
print("Number of Rows       :", df.shape[0])
print("Number of Columns :", df.shape[1])

# Data types
print("\n========== DATA TYPES ==========")
print(df.dtypes)

# Missing values
print("\n========== MISSING VALUES ==========")
print(df.isnull().sum())

# Statistical summary
print("\n========== STATISTICAL SUMMARY ==========")
print(df.describe(include='all'))
```

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```
Saving employee.csv to employee.csv

========== DATASET ==========

    EmpID            Age  Gender  Department  Salary   Experience       City
0                    25.0   Male          HR  35000.0           2    Chennai
1   Name             30.0 Female     Finance  50000.0           5  Bangalore
2     101     Amit   28.0   Male          IT  45000.0           4  Hyderabad
3     102    Priya   35.0 Female   Marketing  60000.0           8
4     103    Rahul   40.0   Male       Sales  70000.0
5     104    Sneha   27.0 Female          IT  48000.0            Mumbai
6     105    Karan   32.0   Male     Finance  55000.0          12      Delhi
7     106    Meen    29.0 Female          HR  42000.0           3    Chennai
8            a       38.0   Male       Sales  68000.0           6       Pune
9     107    Arjun   26.0 Female   Marketing  47000.0           4 Coimbatore
10    108    Kavya    NaN   Male          IT  52000.0          10    Kolkata
11    109   Vikram   31.0 Female     Finance       Na           2  Bangalore
12    110    Divya   45.0   Male   Manageme   N                 5    Chennai
13    111    Manoj   24.0 Female          nt  90000.0           7  Hyderabad
14    112   Anitha   36.0   Male           H  33000.0          20
      113   Rames                          R  61000.0
========== DATASET SHAPE ==========       IT                    Mumbai
Number of Rows    : 15                                          1      Salem
Number of Columns : 8                                           9     Trichy
      114   Keerthi
      115   Suresh

========== DATA TYPES ==========
EmplD                int64
Name               object
Age                float64
Gender              object
Department          object
Salary             float64
Experience           int64
City                object
dtype: object
========== MISSING VALUES ==========
EmpID         0
Name          0
Age           1
Gender        0
Department    0
Salary        1
Experience    0
City          0
dtype: int64

========== STATISTICAL SUMMARY ==========
           EmpID Name        Age Gender Department        Salary  \
count  15.000000   15  14.000000     15         15     14.000000
unique       NaN   15        NaN      2          6           NaN
top          NaN Amit        NaN   Male         IT           NaN
freq         NaN    1        NaN      8          4           NaN
mean  108.000000  NaN  31.857143    NaN        NaN  54000.000000
std     4.472136  NaN   6.187545    NaN        NaN  15104.762367
min   101.000000  NaN  24.000000    NaN        NaN  33000.000000
25%   104.500000  NaN  27.250000    NaN        NaN  45500.000000
50%   108.000000  NaN  30.500000    NaN        NaN  51000.000000
75%   111.500000  NaN  35.750000    NaN        NaN  60750.000000
max   115.000000  NaN  45.000000    NaN        NaN  90000.000000

        Experience     City
count    15.000000       15
unique         NaN       10
top            NaN  Chennai
freq           NaN        3
mean      6.533333      NaN
std       4.882427      NaN
min       1.000000      NaN
25%       3.500000      NaN
50%       5.000000      NaN
75%       8.500000      NaN
max      20.000000      NaN
```

```
import pandas as pd
import matplotlib.pyplot as plt from
google.colab import files

# Upload CSV file
uploaded = files.upload()

# Read uploaded file
filename = list(uploaded.keys())[0] df =
pd.read_csv(filename)

# Display dataset
```

```
print("Student Dataset:")
print(df)

# Calculate frequency distribution of grades
grade_frequency = df['Grade'].value_counts().sort_index()

# Display frequency distribution
print("\nFrequency Distribution of Grades:")
print(grade_frequency)

# Draw Bar Chart
plt.figure(figsize=(6,4))
plt.bar(grade_frequency.index,  grade_frequency.values)

# Chart labels
plt.title("Frequency Distribution of Students' Grades")
plt.xlabel("Grades")
plt.ylabel("Number of Students")

# Display frequency on bars
for i, value in enumerate(grade_frequency.values):
    plt.text(i, value + 0.1, str(value), ha='center')

plt.show()
```

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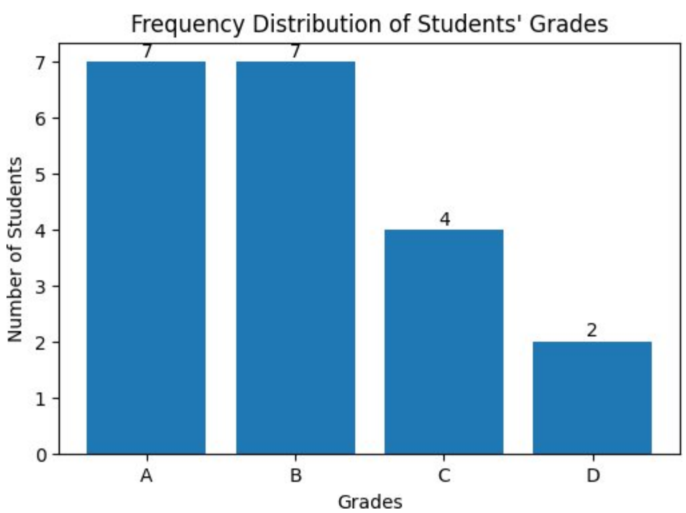
```
Saving student_grades.csv to student_grades.csv
Student Dataset:
    StudentID     Name
0        101      Amit Grade   A
1        102      Priya        B
2        103      Rahul        A
3        104      Sneha        C
4        105      Karan        B
5        106      Meen         A
6                 a            B
7        107      Arjun        C
8        108      Kavya        A
9        109     Vikram        B
10       110      Divya        D
11       111      Manoj        A
12       112     Anitha        B
13       113     Rames         C
14               h             A
15       114   Keerthi         B
16       115     Suresh        A
17       116      Akash        D
18       117      Nisha        C
19       118      Rohit        B
         119      Deep
Frequency Distribution of Grades:
Grade    120      Varun
A    7
B    7
C    4
D    2
Name: count, dtype: int64
```




```
import pandas as pd
import matplotlib.pyplot as plt from
google.colab import files

# Upload CSV file
```

```
uploaded = files.upload()

# Read dataset
filename = list(uploaded.keys())[0] df =
pd.read_csv(filename)

# Display dataset
print("House Price Dataset:")
print(df)

# Plot Histogram
plt.figure(figsize=(7,5))
plt.hist(df['Price'], bins=6)

plt.title("Histogram of House Prices")
plt.xlabel("House Price")
plt.ylabel("Frequency")

plt.show()

# Calculate Skewness
skewness = df['Price'].skew()
print("\nSkewness =", round(skewness, 2)) #

Identify Distribution

if abs(skewness) < 0.5:
    print("The data is approximately Normally Distributed.") elif
skewness > 0:
    print("The data is Positively (Right) Skewed.") else:
    print("The data is Negatively (Left) Skewed.")
```

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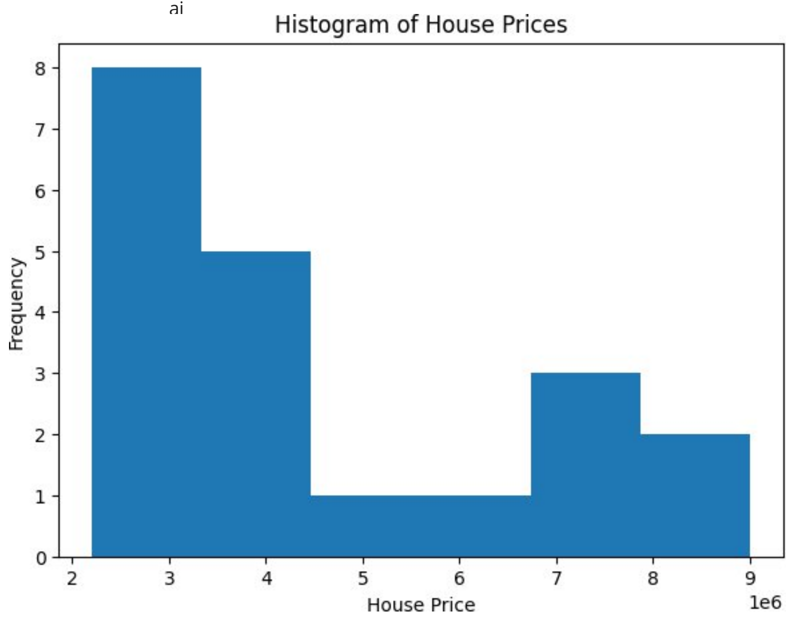
Saving house prices.csv to house prices.csv
House Price Dataset:

	HouseID	Location	Price
0	1	Chennai	2500000
1	2	Bangalore	3200000
2	3	Hyderabad	2800000
3	4	Mumb	7500000
4		ai	6800000
5	5	Delhi	4100000
6	6	Pune	3500000
7	7	Kolkata	2200000
8	8	Coimbatore	2900000
9	9	Chennai	3600000
10	10	Bangalore	3100000
11	11	Hyderabad	8500000
12	12	Mumb	7200000
13		ai	4300000
14	13	Delhi	2600000
15	14	Pune	2700000
16	15	Kochi	3400000
17	16	Chennai	9000000
18	17	Bangalore	6500000
19	18	Mumb	4500000
		ai	



Skewness = 0.9
The data is Positively (Right) Skewed.

```
import pandas as pd
import matplotlib.pyplot as plt from
google.colab import files

# Upload CSV file
uploaded = files.upload()

# Read the uploaded file
filename = list(uploaded.keys())[0] df =
pd.read_csv(filename)

# Display dataset
print("Salary Dataset:")
print(df)

# Create Box Plot
plt.figure(figsize=(6,5))
plt.boxplot(df['Salary'], vert=True)

plt.title("Box Plot of Employee Salaries")
plt.ylabel("Salary")
plt.grid(True)

plt.show()

# Detect Outliers using IQR Method Q1 =
df['Salary'].quantile(0.25)    Q3 =
df['Salary'].quantile(0.75)
IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
```

```
upper_limit = Q3 + 1.5 * IQR
outliers = df[(df['Salary'] < lower_limit) | (df['Salary'] > upper_limit)] print("\nLower Limit:",

lower_limit)

print("Upper Limit:", upper_limit)

if len(outliers) > 0:
    print("\nOutliers Found:")
    print(outliers[['EmpID', 'Name', 'Salary']]) else:
    print("\nNo Outliers Found.")
```

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```
Saving salary_data.csv to salary_data.csv
Salary Dataset:

     EmpID              Department  Salary
0                               HR   35000
1    Name                  Finance   48000
2      101      Amit            IT   52000
3      102     Priya     Marketing   60000
4      103     Rahul         Sales   45000
5      104     Sneha            IT   50000
6      105     Karan       Finance   55000
7      106      Meen            HR   42000
8                  a         Sales   70000
9      107     Arjun     Marketing   58000
10     108     Kavya            IT   53000
11     109    Vikram       Finance   49000
12     110     Divya     Managemen  150000
13     111     Manoj             t   41000
14     112    Anitha             H   51000
15     113     Rames             R   47000
16                 h            IT   59000
17     114   Keerthi         Sales   52000
18     115    Suresh     Marketing   43000
19     116     Akash       Finance   48000
       117     Nisha             H
```


```
Lower Limit: 32625.0
Upper Limit: 69625.0

Outliers Found:
     EmpID          Salary
8                    70000
12   Name           150000
       109  Vikram
```

```
import pandas as pd
from google.colab import files

# Upload CSV file
uploaded = files.upload()

# Read uploaded file
filename = list(uploaded.keys())[0] df =
pd.read_csv(filename)

# Display dataset
print("Students Marks Dataset:")
print(df)
```

```
# Calculate Quartiles
Q1 = df['Marks'].quantile(0.25) Q3 =
df['Marks'].quantile(0.75)

# Calculate IQR
IQR = Q3 - Q1

# Calculate lower and upper limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("\nQ1 :", Q1)
print("Q3 :", Q3)
print("IQR:", IQR)
print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)

# Detect Outliers
outliers = df[(df['Marks'] < lower_limit) | (df['Marks'] > upper_limit)]

if len(outliers) > 0:
    print("\nOutliers Found:")
    print(outliers)
else:
    print("\nNo Outliers Found.")
```

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```
Saving student_marks.csv to student_marks.csv
Students Marks Dataset:
     StudentID      Name
0    Marks
1          101      Amit    78
2          102     Priya    85
3          103     Rahul    67
4          104     Sneha    92
5          105     Karan    74
6          106     Meena    88
7          107     Arjun    81
8          108     Kavya    76
9          109    Vikram    95
10         110     Divya    69
11         111     Manoj    84
12         112    Anitha    73
13         113    Ramesh    90
14         114   Keerthi    79
15         115    Suresh    82
16         116     Akash    77
17         117     Nisha    86
18         118     Rohit    71
19         119     Deepa    80
           120     Varun    25
Q1 : 73.75
Q3 : 85.25
IQR: 11.5
Lower Limit: 56.5
Upper Limit: 102.5

Outliers Found:
    StudentID    Name  Marks
19        120   Varun     25
```

```
import pandas as pd
import matplotlib.pyplot as plt from
google.colab import files

# Upload CSV file
uploaded = files.upload()

# Read dataset
filename = list(uploaded.keys())[0] df =
pd.read_csv(filename)

print("Original Dataset:")
print(df)

# ----------------------------
# Plot Histogram Before Removing Outliers #
plt.figure(figsize=(6,4))------------
plt.hist(df['Score'], bins=8)
plt.title("Histogram Before Removing Outliers")
plt.xlabel("Score")
plt.ylabel("Frequency")
plt.show()
```

```
# -----------------------------
# Plot Box Plot Before Removing Outliers #
plt.figure(figsize=(4,5)) ------------
plt.boxplot(df['Score'])
plt.title("Box Plot Before Removing Outliers")
plt.ylabel("Score")
plt.show()

#
# Remove Outliers using IQR # -----
Q1 = df['Score'].quantile(0.25) Q3 =
df['Score'].quantile(0.75) ----------
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
df_clean = df[(df['Score'] >= lower) & (df['Score'] <= upper)]

print("\nDataset After Removing Outliers:")

print(df_clean)

#
# Histogram After Removing Outliers #
plt.figure(figsize=(6,4)) ------------
plt.hist(df_clean['Score'],  bins=8)
plt.title("Histogram After Removing Outliers")
plt.xlabel("Score")
plt.ylabel("Frequency")
plt.show()

#
# Box Plot After Removing Outliers #
plt.figure(figsize=(4,5))
plt.boxplot(df_clean['Score']) -------
plt.title("Box Plot After Removing Outliers")
plt.ylabel("Score") ----------------
plt.show()
```